

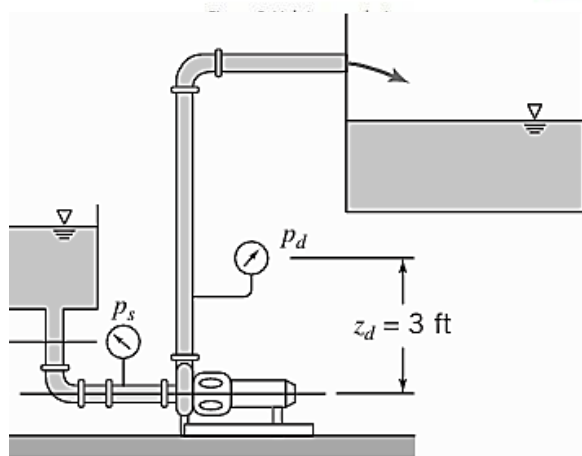
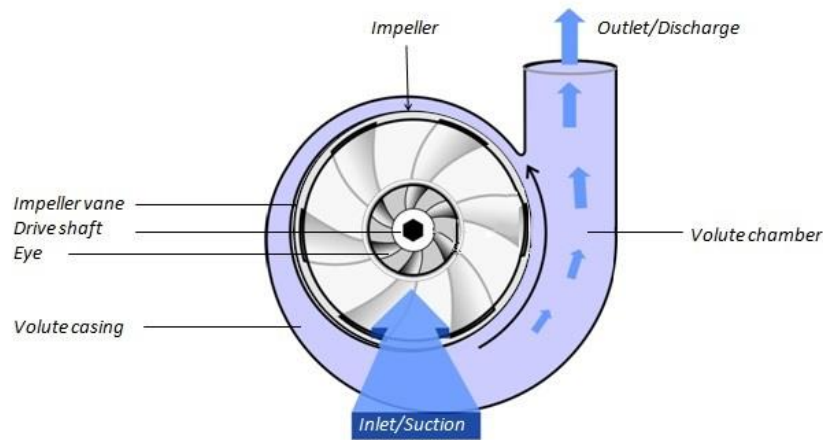


Final Project of Fluid Mechanics I
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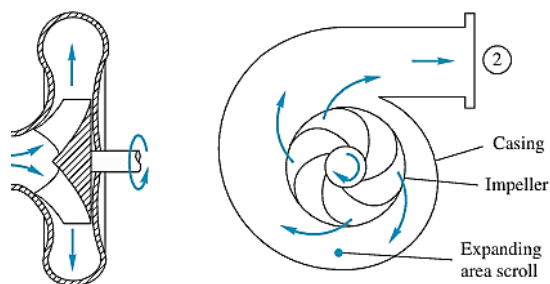
PROJECT TITLE: CENTRIFUGAL PUMP

INTRODUCTION

Centrifugal pumps are among the most widely used fluid transfer equipment in various engineering industries. Understanding their operating principles and performance characteristics plays a crucial role in the design and operation of fluid transport systems. This project is designed to familiarize students with the working principles of centrifugal pumps, analyze performance curves, select appropriate pumps, and examine operational issues.



B



A

In Figure (A), a schematic view of the main components of a centrifugal pump is shown. On the left side of Figure (A) (labeled 1), the fluid flow path inside the impeller is illustrated, where the rotating impeller imparts kinetic energy to the fluid. On the right side (labeled 2), a top view of the pump is presented, showing the main components, including the impeller, the volute casing, and the spiral region with an expanding cross-sectional area. This illustration clearly demonstrates the process of converting mechanical energy into pressure energy in centrifugal pumps and is very useful for understanding their operating principles.

Figure (B) shows a simple system containing a centrifugal pump responsible for transferring fluid from a source tank to a destination tank.

Based on the explanations provided, carefully study the following case studies and complete the required tasks accurately, then report your results. The case studies consist of three main parts, all of which are mandatory for students to complete.

Case Studies

1)

- a) Considering an appropriate control volume around the pump, derive the governing equations (conservation equations).
- b) For the performance parameters listed below, provide a complete description based on research from various sources and express their relationships in terms of measurable pump parameters (density, volumetric flow rate, torque, rotational speed):

- **Hydraulic power**
- **Mechanical power**
- **Pump head**
- **Specific speed**
- **Hydraulic efficiency of the pump**
- **Overall (total) efficiency of the pump**

2-

a) Obtain the required parameters and plots and analyze them: (You may perform calculations by hand or using software such as EES, Excel, MATLAB, or Python.)

In this section, considering the flow system in Figure (b) which is used to test a centrifugal pump at a specified nominal speed, two different data sets are provided; using these you must obtain the requested items:

Dataset 1:

- **Nominal speed: 1750 rpm**
- **Water temperature: 80 °F**
- **Suction and discharge pipe diameters: 6 in**
- **The elevations are given in Figure (b).**
- **The electric motor is supplied by a three-phase 460 V line, its power factor is 0.875 and its fixed efficiency is 90%. (Hint: mechanical efficiency is 100% and the electrical output power of the motor equals the mechanical input power to the pump; for a three-phase motor the power can be calculated using the following relation:**

$$P_{in} = \eta\sqrt{3} \times (\text{power factor}) \times \text{Voltage}$$

The measured data are as follows:

Flow rate (gpm)	Suction pressure (psig)	Discharge pressure (psig)	Motor Current (amp)
0	0.65	53.3	18.0
500	0.25	48.3	26.2
800	-0.35	42.3	31.0
1000	-0.92	36.9	33.9
1100	-1.24	33.0	35.2
1200	-1.62	27.8	36.3
1400	-2.42	15.3	38.0
1500	-2.89	7.3	39.0

2-1- Using the given data table, determine the pump head, hydraulic power, and overall pump efficiency at a flow rate of 800 gpm.

2-2- For the provided dataset, plot the pump head, hydraulic power, specific speed, and overall pump efficiency as functions of volumetric flow rate and analyze the behavior of each curve. Attempt to fit a polynomial to the obtained curves (you may perform the curve fitting with Excel, MATLAB, or any other programming language).

Dataset 2:

- Nominal speed: 1100 rpm
- Water temperature: 25 °C
- Suction and discharge pipe diameters: 15 cm
- Unlike dataset 1, neglect the elevation difference.
- The measured data are as follows: (Note: in this section the motor power is reported instead of the motor current and there is no need to recalculate it. Also note that you must calculate the mechanical power here as requested in the problem statement below.)

Torque (N.m)	Flow rate (L/min)	Suction pressure (bar)	Discharge pressure (bar)	Motor Power (kW)
2.1	254	-0.08	0.06	0.52
2	228	-0.07	0.11	0.5
1.9	197	-0.05	0.18	0.48
1.7	163	-0.04	0.25	0.46
1.8	177	-0.05	0.21	0.47
1.7	155	-0.04	0.25	0.45
1.5	127	-0.03	0.29	0.42
1.5	129	-0.03	0.29	0.42
1.4	99	-0.02	0.32	0.4
1.2	75	-0.02	0.34	0.38
1.1	50	-0.02	0.35	0.36
1	27	-0.01	0.36	0.33
0.9	2	-0.01	0.36	0.31

2-3- Using the given data table, determine the pump head, hydraulic power, hydraulic efficiency, and overall pump efficiency at a flow rate of 197 L/min.

2-4- For the provided dataset, plot pump head, hydraulic power, mechanical power, specific speed, and overall pump efficiency as functions of volumetric flow rate and analyze the behavior of each curve. Try to fit a polynomial to the obtained curves (you may perform the curve fitting with Excel, MATLAB, or any other programming language).

b) Selecting a Suitable Pump Using Coding:

In this section, the goal is to write a simple MATLAB or Python code that can recommend an appropriate centrifugal pump model from the *Pompiran* company's catalog based on the operating conditions specified by the user.

The inputs and outputs are as follows:

Input Specifications:

- Volumetric flow rate: in cubic meters per hour (m^3/h)
- Required head: in meters (m)

Output Specifications:

- Suitable pump model (from the families shown on page 5 of the catalog, only for a motor speed of 2900 rpm)
- Impeller diameter
- Pump efficiency at the operating point
- Required electric motor power

Limiting Conditions (an additional score will be given if the program is extended to handle higher flow rates and heads):

- Flow rate less than $40 \text{ m}^3/\text{h}$
- Head less than 40 m

Important Notes for Coding:

The data corresponding to the pump family curves (overlapping curves) and the head–flow (H–Q) characteristics (based on impeller diameter), efficiency, and power charts must be extracted from the catalog diagrams using digitizer software and then used in the program as a data file or matrix.

To determine the efficiency, use interpolation between the points extracted from the constant-efficiency curves.